

Nithin Jambula

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PROFESSIONAL EXPERIENCE

- Ananta Technologies** Dec 2026 – Present
Robotics Software Intern India
 - Collaborated with US-based client to define solution designs, enhance build test efficiency, and streamline simulation workflows, reducing validation time by **35%**.
 - Developed and executed **50+** robotic test scenarios using MuJoCo simulator for performance validation and acceptance testing.
 - Performed real-world testing on Webots robotic platforms and analyzed system behavior for quality assurance.
 - Successfully supported deployment of **2 production robots** within the internship timeline.
 - Facilitated cross-team communication between onshore and offshore development teams, ensuring clear articulation of needs.
- AIR Centre VIT-AP** [!\[\]\(38441ceaa711016e0bf2ad46ad394ff4_img.jpg\)](#) Jun 2024 to Nov 2025
Full Stack Engineer / Student Researcher Amaravathi, India
 - Architected and implemented **scalable microservices** for neuro-symbolic AI security testing platform with PostgreSQL-backed storage.
 - Developed distributed testing infrastructure with database replication, failover automation (Patroni/PgBouncer), and high availability.
 - Built and deployed containerized services using Docker with Kubernetes orchestration for horizontal scaling.
 - Implemented monitoring (Prometheus/Grafana) and infrastructure-as-code (Terraform) for reproducible environments.
 - Led code reviews, enforced best development practices, and mentored junior developers on API security and testing.
 - Built automated security validation pipelines for API vulnerability detection (SQLi, XSS, BOLA), improving discovery by **34%**.
 - Managed competing priorities and provided status updates to leadership on project milestones.
- Vulcan: Self-Driving EV (VIT-AP Initiative)** Jan 2023 – Present
Software Engineer / Team Lead Amaravathi, India
 - Led perception and navigation software development for a self-driving campus EV.
 - Trained DeepLabV3+ on Mapillary data, achieving **67%** multi-class road scene segmentation accuracy.
 - Implemented lane-following and turn-handling logic for controlled campus environments.
 - Collaborated with cross-functional teams including hardware, controls, and systems engineers.

KEY PROJECTS

- 16fps: Multi-Agent Video Generation Platform (SaaS)** Oct 2025 – Present
Python • LangChain • Stable Video Diffusion • Docker • Kubernetes • GCP [!\[\]\(30a147af384f9f71632c2ff17bc706c8_img.jpg\)](#)
 - Built a **scalable SaaS platform** with autonomous multi-agent orchestration for video generation workflows.
 - Designed self-correcting agent pipeline with memory, tool-use, and reflection for consistent output.
 - Enabled parallel multi-video rendering with dynamic style transfer, deployed as a scalable service.
 - Sold commercial rights and operational access to a media production company.
- EchoSight: Assistive Navigation Glasses (Patent Pending)** Dec 2024 - Present
YOLOv8 • Roboflow • Raspberry Pi • GCP • OpenCV • TTS [!\[\]\(9b33568d5c136f08ca688ce48be37574_img.jpg\)](#)
 - Developed a wearable vision-based system for real-time object detection and navigation feedback.
 - Fine-tuned YOLOv8 with custom dataset (50+ object classes), enabling **45+ FPS** detection.
 - Integrated spatial direction and distance estimation with contextual alerting system.
- WOAT: Context-Aware Conversational Agent** Jul 2023 – Nov 2023
Gemini API • GCP • Agentic Workflow • Python • Node.js [!\[\]\(8c93063dab026f10e159986b27c41c64_img.jpg\)](#)

- Developed an agentic conversational system understanding user intent, tone, and context.
- Designed agentic reasoning pipeline for analyzing chat history and inferring relationship dynamics.
- Implemented WhatsApp integration layer with context memory and secure login.

• **PostgreSQL High-Availability Cluster**

Jun 2024 – Present

PostgreSQL • Patroni • PgBouncer • Terraform • Ansible • Grafana



- Automated PostgreSQL cluster deployment using Terraform/Ansible with streaming replication.
- Implemented automatic failover using Patroni (**99.9% uptime**) with PgBouncer connection pooling.
- Built Grafana dashboards for replication lag and query performance monitoring.

TECHNICAL SKILLS

- **Front-End** React, JavaScript, TypeScript, HTML5, CSS3, Next.js, Vite.js, REST APIs, Responsive Design
- **Back-End & Microservices** Python, Node.js, FastAPI, Flask, RESTful APIs, GraphQL, Microservices Architecture, API Design
- **Databases** PostgreSQL, Redis, TimescaleDB, Schema Design, Query Optimization, Indexing, Patroni, PgBouncer
- **Cloud & DevOps** Docker, Kubernetes, Terraform, Ansible, GCP, AWS (EC2, S3, RDS), CI/CD (GitHub Actions), Linux
- **Observability & Security** Prometheus, Grafana, ELK Stack, OWASP, Security Testing, Vulnerability Assessment, Deepfake Detection
- **AI/ML & Robotics** PyTorch, TensorFlow, YOLOv8, DeepLabV3+, LangChain, ROS, Gazebo, MuJoCo, Sim-to-Real, NVIDIA Jetson
- **Soft Skills** Code Reviews, Mentoring, Cross-Team Collaboration, Agile/Scrum, Problem Solving, Analytical Thinking, Attention to Detail

PUBLICATIONS & PATENTS

- **Patent Pending:** EchoSight — AI-Powered Wearable Navigation System
- **Neuro-Symbolic AI for Adaptive API Security Testing** — AIRC 2026 DOI: 10.1109/AIRC69745.2026.1163138
- **AI-Driven Detection of Deepfakes & Counterfeit Currency** — ISAI-NLP 2025 DOI: 10.1109/ISAI-NLP66160.2025.1

ACHIEVEMENTS

- **Winner — Engineering Clinics 2024:** First place for real-time threat detection system
- **Awardee — Vikas 2024 Innovation Challenge:** Recognized for EchoSight assistive technology
- **Ananta Technologies:** Successfully deployed two production robots within internship timeline

EDUCATION & CERTIFICATIONS

B.Tech CSE	VIT-AP University, Amaravathi	8.0/10 (2023–2027)
Class XII	Narayana Junior College	95% (2021–2023)
Class X	Vivekananda School	99.3% (2021)
Certifications: PostgreSQL Administration • Docker/Kubernetes • Stanford ML • IBM Deep Learning • Google Object Detection		